

TO: Honorable Mayor and City Council

FROM: Joseph La Margo, City Manager



SUBJECT: Elijah Root Dam Removal Feasibility Study

SUPPORTING PERSONNEL: Kathleen Hoyle, Director of Parks, Recreation & Senior Citizen Services
Kendra Gwin, Director of Transportation & Utilities
Jamie Harmon, Assistant City Engineer

ACTION RECOMMENDED: Award a contract in the amount of \$119,805 to Wightman for engineering consulting services for the Elijah Root Dam Removal Feasibility Study, and authorize the City Manager to execute all documents related to this action on behalf of the city.

The Elijah Root Mill Dam was part of a lumber mill originally constructed in 1834. The lumber mill structure has been removed but the dam structure including an earthen berm and two concrete spillways remain that split the flow of Portage Creek around an island located downstream of the dam. Wetlands exist upstream of the dam and a significant amount of sediment has accumulated behind the dam. The East Milham Avenue bridge is upstream of the dam and the Lovers Lane access road is downstream. The dam is failing and is a barrier for fish passage and recreational uses including canoeing and kayaking. Therefore, the City Administration believes removal of the dam is prudent and recommends the preparation of a feasibility study for the removal of the Elijah Root Mill Dam.

There are three major engineering elements which are critical to the removal of the dam:

1. Scour prevention of the Milham Avenue bridge footings from the water level drop.
2. Management of more than seven feet of sediment behind the dam.
3. Overall hydrology of Portage Creek as part of the restoration with impact to the Lovers Lane access road.

These elements are a major focus of an assessment to prevent bridge failures and sediment buildup that could potentially force Portage Creek to reroute into the pedestrian trails. Experience in dam removal, sediment management, hydrology and bridge impact assessments is a major factor in the vendor selection for the feasibility study. The city is also looking at qualifications for creek restoration that will allow for streambed restoration for brown trout and other marine life, recreational usage and historic

interpretation.

The goal of the feasibility study would be to evaluate alternatives and define the scope and sequencing of work that would be necessary to remove the Elijah Root Mill Dam, while minimizing impacts to the East Milham Avenue Bridge, Lovers Lane access road and adjacent wetland areas. The feasibility study will NOT include design engineering services such as preparing project plans and specifications and securing permits. Upon completion of the feasibility study, the city would consider negotiating engineering services for the safe removal of the Elijah Root Mill Dam. The final report with recommendations will include:

- Identify and evaluate three alternatives to allow for bridge integrity, kayak/canoe and fish passage
- Identify and evaluate necessary alterations to upstream and downstream creek morphology, including the creek bed and streambank stabilization, as well as restoration and floodplain alterations
- Assessment and recommendation of the Lover's Lane culvert to allow unobstructed flow of Portage Creek and elimination of sediment deposits
- Maintain portions of the mill building structure for historical interpretation
- Restoration of fish habitat and streambed
- Cost estimates for all alternatives

Proposals were solicited for conducting the Elijah Root Mill Dam Removal Feasibility Study. Four proposals were received on December 18, 2020 to perform the work. The firm that submitted the low bid is not qualified and does not have the high level of experience with dam removals and bridge assessments held by the other bidding firms. Wightman submitted the highest bid in the amount of \$119,805 (within \$2,135 of the other two proposals). City staff have worked extensively with Wightman, which has experience and familiarity conducting the city's bridge inspections every two years, including the East Milham Avenue Bridge. The firm has three dedicated engineers with dam removal experience and has MDOT hydraulic analysis and modeling qualifications, which would be required for this project. Wightman has an excellent and highly experienced team in place for stream restoration and historic interpretation proposed for the project.

It is recommended that City Council award a contract in the amount of \$119,805 to Wightman for engineering consulting services for the Elijah Root Mill Dam Removal Feasibility Study, and authorize the City Manager to execute all documents related to this action on behalf of the city.

FUNDING: Funding is included in the CIP for FY 19-20.

Attachments: 1. Proposal Tabulation - Elijah Root Dam

**ELIJAH ROOT DAM REMOVAL FEASIBILITY STUDY
PROPOSAL TABULATION**

	Total Sum
Land & Resource Engineering	\$99,340.00
Keiser & Associates, LLC	\$117,670.00
GEI Consultants of Michigan, P.C.	\$118,500.00
Wightman	\$119,805.00